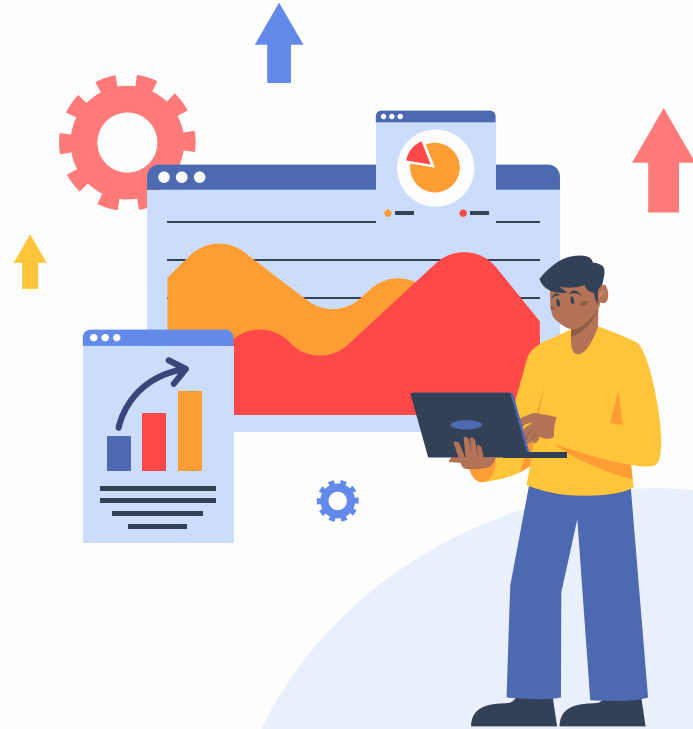
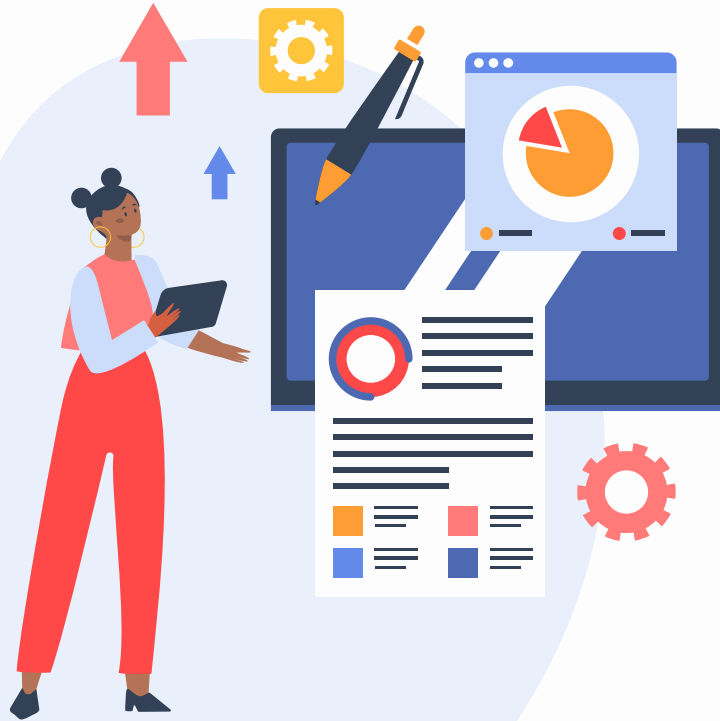


Indicators of Income Inequality in the US

Michelle Cameron, Valentina Hong, Ian Howard





Guiding Questio

n

How does socioeconomic inequality vary across geographic areas, and what factors are most strongly associated with it?



Motivation

- We wanted to explore patterns of socioeconomic inequality across the US
- Socioeconomic inequality has become a more divisive issue in recent years
- Specifically exploring the geographic distributions and factors affecting socioeconomic inequality





Subquestions

Inequality Clusters

Are there clusters of high inequality, and how have they changed over time?

Employment Patterns

How is income inequality manifested in unemployment patterns across the United States?

Education and Income

Is there a relation between educational attainment and income and where is most affected?

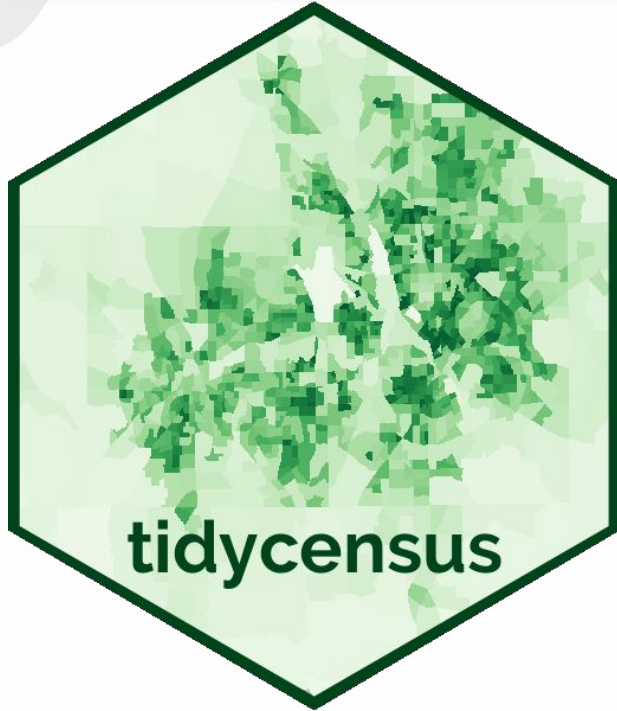


Dataset: tidycensus

`tidycensus` is an R package that allows users to query data from the US Census Bureau's data APIs

We used it to query the following data:

- **Inequality clusters:** Maryland county-level Gini index data, which was then used to calculate LISA clusters
- **Employment patterns:** US county-level employment status data, used 5-year ACS to get counties with smaller populations (leaflet used for interactive map)
- **Education and income:** US state level median household income, poverty rates, and education level (as measured by % with degree, % with high school diploma)



Demo!





01 Inequality Clusters

QUESTION

Are there clusters of high inequality, and how have they changed over time?

MAP SETTINGS

Map Year

2010

2024

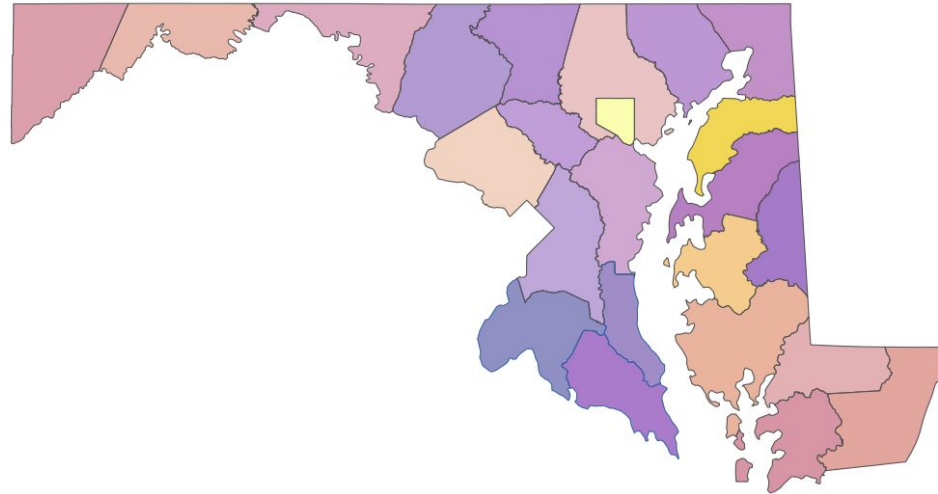


Fill counties using

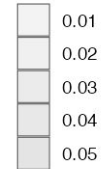
- ☒ Gini index
- ☐ LISA cluster type

Load Data

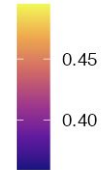
County Inequality in Maryland



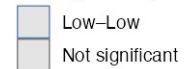
Gini MOE



gini



LISA (border)

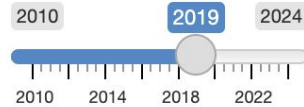


QUESTION

Are there clusters of high inequality, and how have they changed over time?

MAP SETTINGS

Map Year

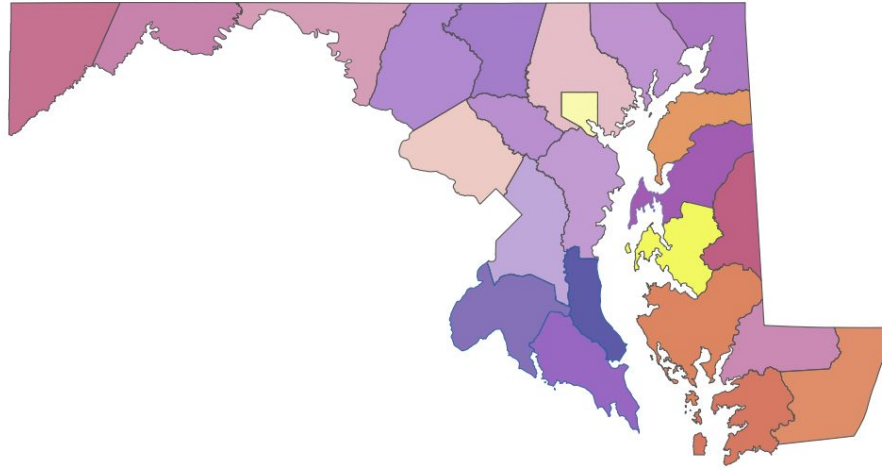


Fill counties using

- ☒ Gini index
- ☐ LISA cluster type

Load Data

County Inequality in Maryland



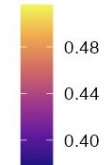
LISA (border)

- ☐ Low-High (outlier)
- ☐ Low-Low
- ☐ Not significant

Gini MOE

- ☐ 0.005
- ☐ 0.010
- ☐ 0.015
- ☐ 0.020
- ☐ 0.025

gini

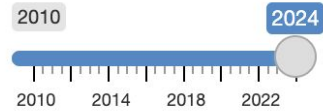


QUESTION

Are there clusters of high inequality, and how have they changed over time?

MAP SETTINGS

Map Year

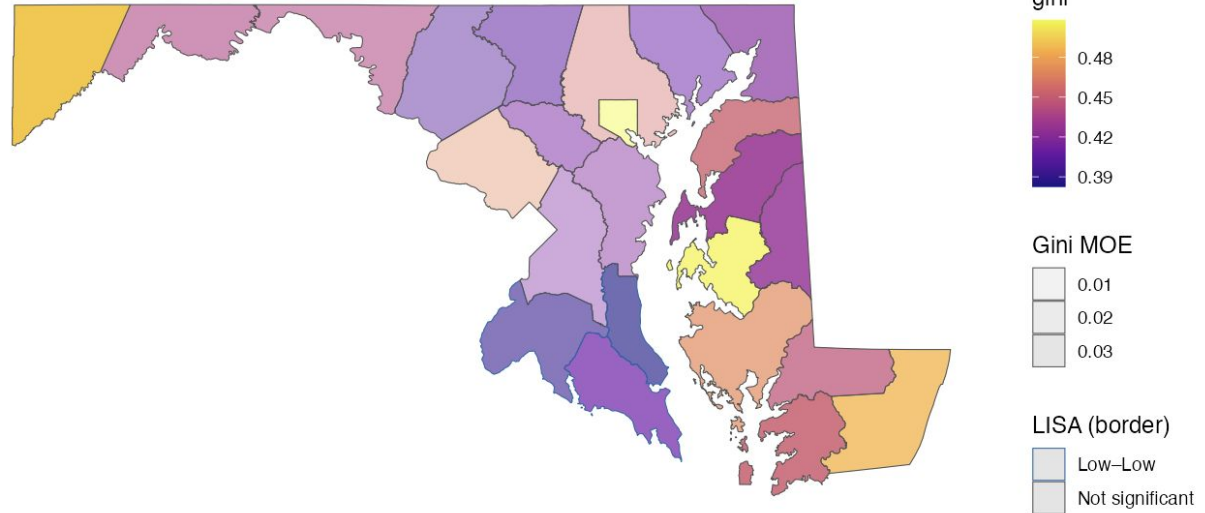


Fill counties using

- ☒ Gini index
- ☐ LISA cluster type

Load Data

County Inequality in Maryland





02

Employment patterns

QUESTION

What geographic patterns exist in unemployment rates across the United States?

CONTROLS

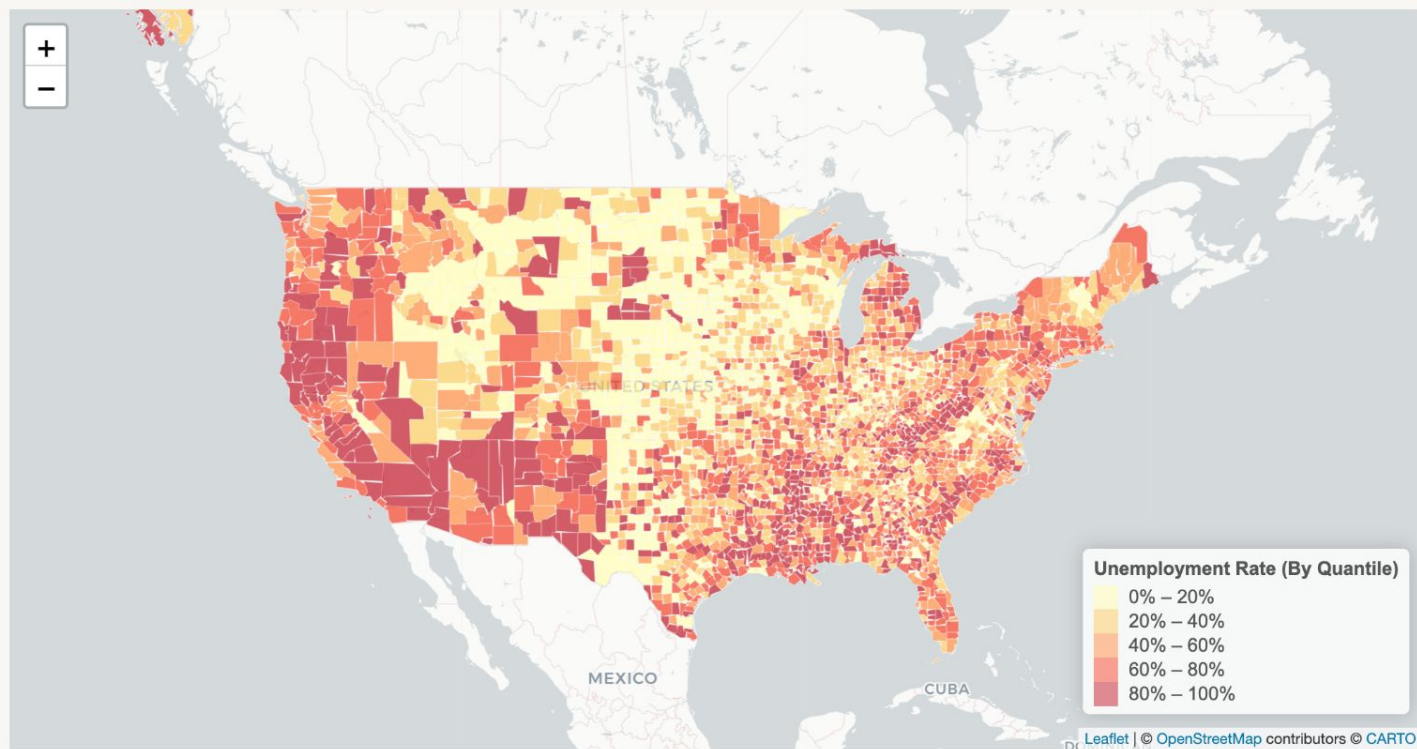
Select a variable to map

Unemployment Rate

Select Year

2015

2022



Highest

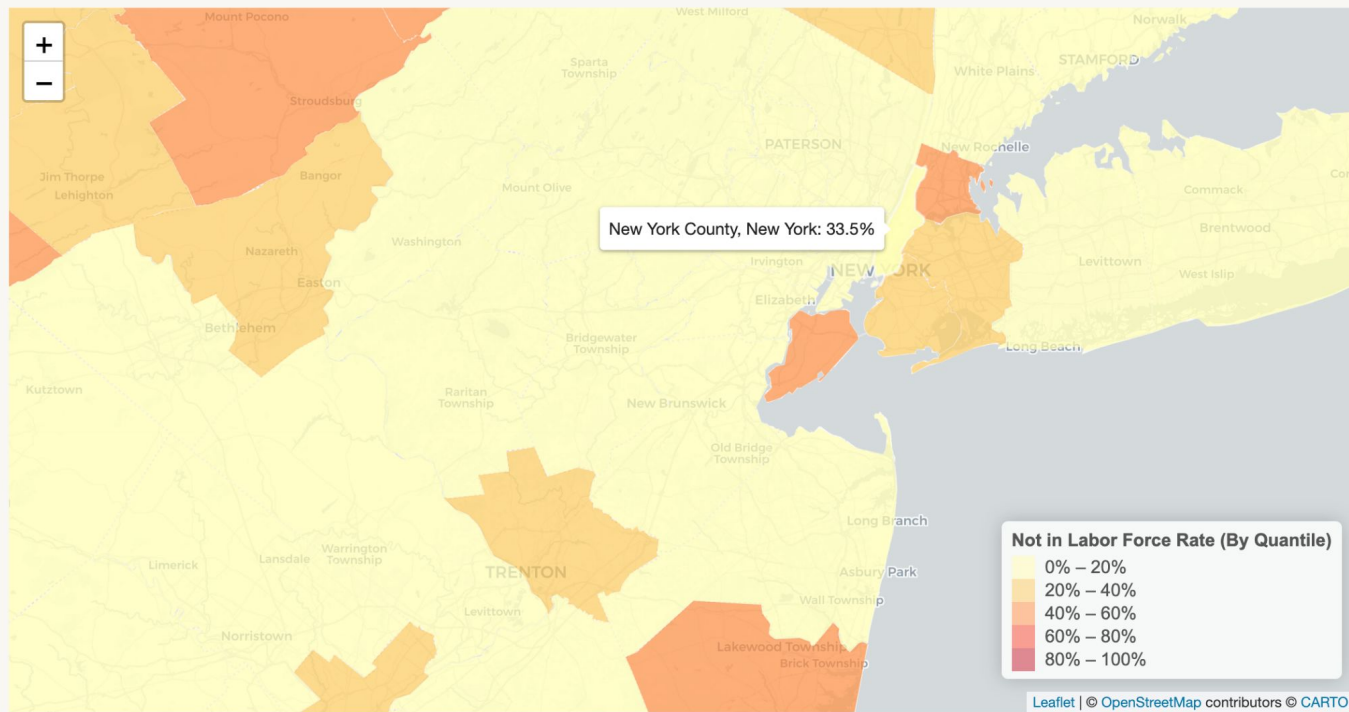
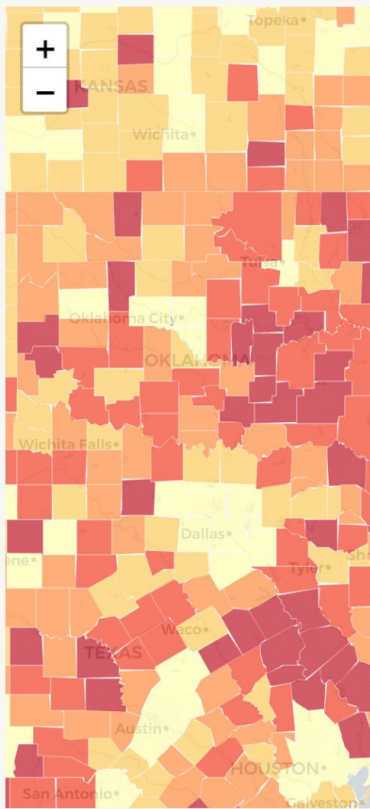
Ziebach County, South Dakota — 32%

Lowest

Kiowa County, Colorado — 0% Kalawao County, Hawaii — 0% Hamilton County, Kansas — 0% Fallon County, Montana — 0% Garfield County, Montana — 0% Wibaux County, Montana — 0% Loup County, Nebraska — 0% McPherson County,

US Median

4.7%





03

Education and Income

QUESTION

Is there a relation between educational attainment levels and poverty rates or median household income

CONTROLS

ACS Year

2022

Y-Axis

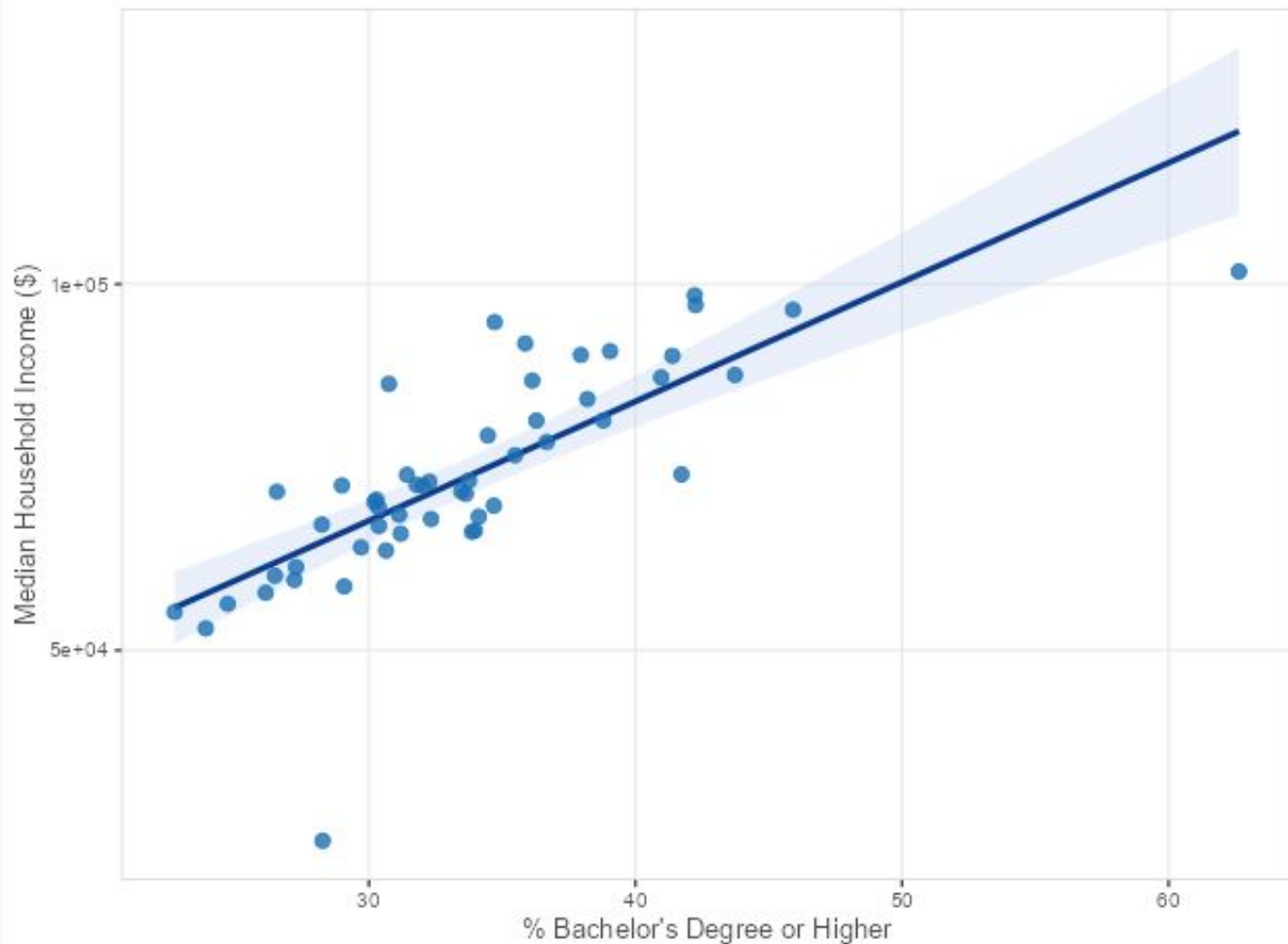
Median Household
Income (\$)

X-Axis (Education)

% Bachelor's
Degree or Higher

Fetch Data

Median Household Income (\$) vs. % Bachelor's Degree or Higher



Source: ACS 5-Year 2022

QUESTION

Is there a relation between educational attainment levels and poverty rates or median household income

CONTROLS

ACS Year

2022

Y-Axis

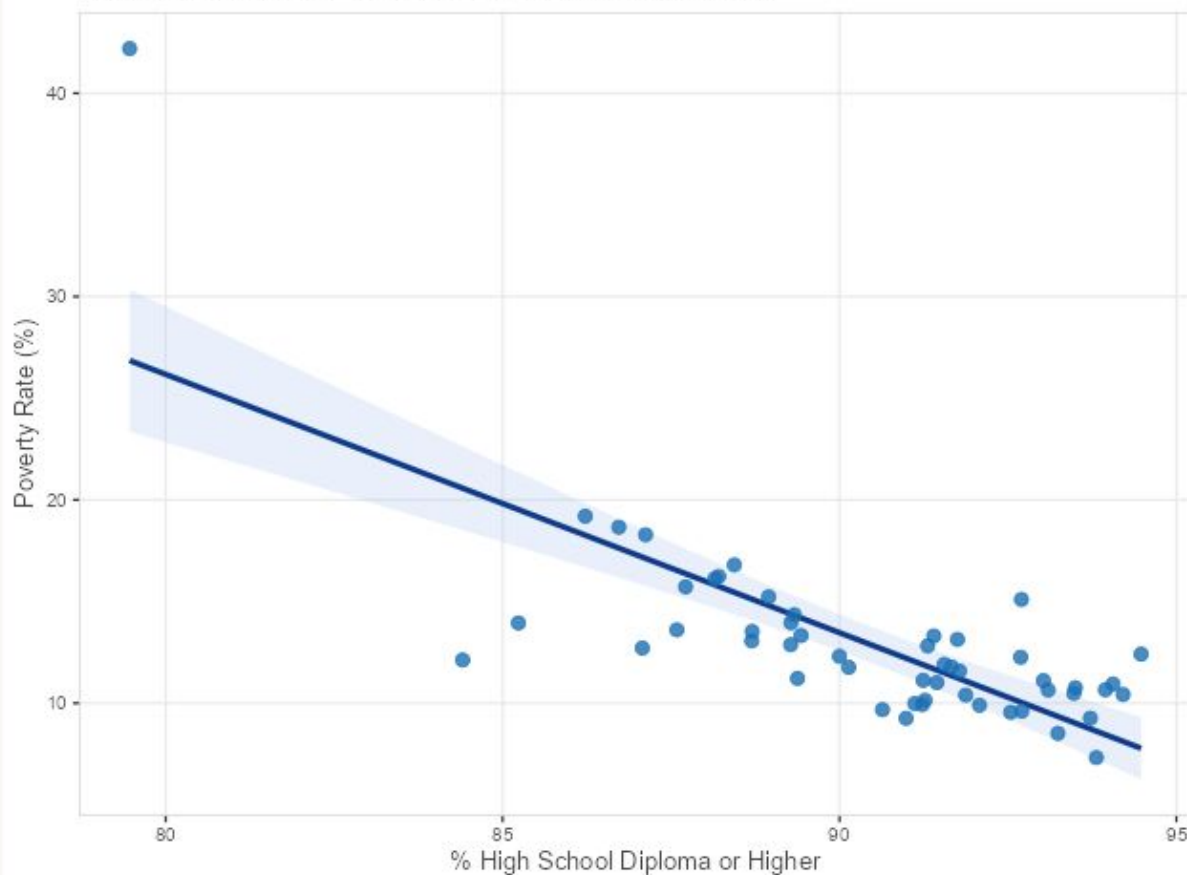
Poverty Rate (%)

X-Axis (Education)

% High School Diploma or Higher

Fetch Data

Poverty Rate (%) vs. % High School Diploma or Higher



Source: ACS 5-Year 2022

Louisiana | % High School Diploma or Higher: 86.7% | Poverty Rate (%): 18.7%

Thanks!

CREDITS: This presentation template was created by [Slidesgo](#), and includes icons by [Flaticon](#), and infographics & images by [Freepik](#)

